Subset Selection and Shrinkage

MATH 4432 Statistical Machine Learning · Tutorial T09

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Choosing a linear model

Many available predictors. Limited information in the data.

Aim	What can go wrong?
Predict new responses	Extra coefficients can fit sampling noise
Explain a fitted model	Many weak or redundant inputs obscure it

Subset selection

Choose which coefficients may be nonzero.

Shrinkage

Control how large the coefficients can be.

A subset defines the candidate model

n observations; p available predictors; a selected set S of size k.

$$y_i = \beta_0 + \sum_{j \in S} x_{ij}\beta_j + \varepsilon_i, \qquad i = 1, \ldots, n$$

Included predictors: estimate their coefficients by least squares.
Excluded predictors: set their coefficients to zero.

$$k = 0: \quad \widehat{y}_i = \overline{y} \qquad \text{(the null model)}$$

The intercept is always included; k counts slopes only.

Comparing fits on the same data

A fitted value predicts an observed response; a residual is its error.

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2$$

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad \mathrm{TSS} = \sum_{i=1}^{n} (y_i - \overline{y})^2$$

Lower RSS = higher R^2, when the response data are unchanged.

Both describe training fit. Neither directly measures future error.

Why training fit favors larger models

Suppose a model uses predictors 1 and 2. Add predictor 3.

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x_1 + \widehat{\beta}_2 x_2 + 0\, x_3$$

The old fit remains available: choose the new coefficient as zero.
Re-optimizing can only lower RSS or leave it unchanged.

The improvement may fit signal, sampling noise, or both.

So training RSS alone cannot choose the model size.

Best subset selection

Stage	Operation
For each size k	Fit every subset containing k predictors
Keep one model at size k	Choose the subset with the smallest RSS
Across sizes 0, ..., p	Choose k using validation or a complexity penalty

The best fit at a fixed size is only a candidate for the final model.

All coefficients, including the intercept, are fitted separately in every candidate.

A complete three-predictor example

Eight observations; three predictors; least-squares refitting each time.

Size k	Subset(s)	RSS
0	None	18
1	{1} {2} {3}	12 12 10
2	{1, 2} {1, 3} {2, 3}	6 9 9
3	{1, 2, 3}	6

Constructed example, not a performance benchmark. Construction and verification: Appendix A2.

The best subsets need not be nested

Size	Winning subset	RSS
0	None	18
1	{3}	10
2	{1, 2}	6
3	{1, 2, 3}	6

The best pair excludes the best single predictor.

A good variable on its own need not belong to the best combination.

Choosing variables and choosing size

At a fixed size k

Search among competing subsets; keep the smallest training RSS.

Across different sizes

Account for extra flexibility: use an information criterion or CV.

First: how can we charge a larger model for its extra flexibility?

AIC and BIC: fit plus a parameter cost

Likelihood: how well a model explains the observed responses.
Larger maximized log-likelihood means better fit.

$$\mathrm{AIC} = -2\hat{\ell} + 2d$$

$$\mathrm{BIC} = -2\hat{\ell} + d\log n$$

d = number of estimated parameters. Choose the smaller score.

AIC emphasizes predictive fit; BIC charges more per parameter when $n > e^2$. [1,2]

Does one more coefficient earn its cost?

100 observations. One added slope improves −2 log-likelihood by 3.

Criterion	Fit change	Extra cost	Score change
AIC	−3	2	−1: prefer larger
BIC	−3	log(100) ≈ 4.61	+1.61: prefer smaller

Same fit improvement, different willingness to pay for complexity.

AIC and BIC for Gaussian linear regression

Assume independent Gaussian errors with a common unknown variance.

$$\mathrm{AIC} = n\log(\mathrm{RSS}/n) + 2k + C_A$$

$$\mathrm{BIC} = n\log(\mathrm{RSS}/n) + k\log n + C_B$$

k slopes + intercept + error variance: d = k + 2.
The constants do not change when comparing these subsets.

Same observations and response required. Derivation: A3. Related Cp / adjusted R^2: A4.

Exhaustive search becomes expensive

$$\sum_{k=0}^{p} \binom{p}{k} = 2^p$$

Stepwise selection reduces search by committing to a path.

p	All subsets	Forward candidates*
5	32	16
10	1,024	56
20	1,048,576	211

*Including the null model: 1 + p(p + 1)/2. Fits can differ in computational cost.

Forward stepwise selection

Step	Action
Start	Fit the intercept-only model
Add	Try each unused predictor; refit all included coefficients
Choose	Keep the addition giving the smallest RSS
Continue	Build one candidate at each attainable size

An included predictor stays on the path. Choose the final size later.

Can start when p > n; an unregularized path still faces rank / sample-size limits.

Forward selection can miss the best pair

Current model	Possible next models	Decision
None	{1}: 12 {2}: 12 {3}: 10	Choose {3}
{3}	{1, 3}: 9 {2, 3}: 9	Choose {1, 3}*
{1, 3}	{1, 2, 3}: 6	Add predictor 2

Best pair: {1, 2}, RSS = 6. Forward pair: {1, 3}, RSS = 9.

The first decision excludes a better pair from the search path.

Entries show training RSS. *Ties use the smaller predictor index.

Backward stepwise selection

Step	Action
Start	Fit the full model with all p predictors
Remove	Try each one-variable deletion; refit the remaining model
Choose	Keep the deletion giving the smallest RSS
Continue	Build one candidate at each smaller size

A removed predictor cannot return. Choose the final size later.

Ordinary full-model fitting requires an estimable design; typically $n > p + 1$ and full rank.

Backward selection can miss the best singleton

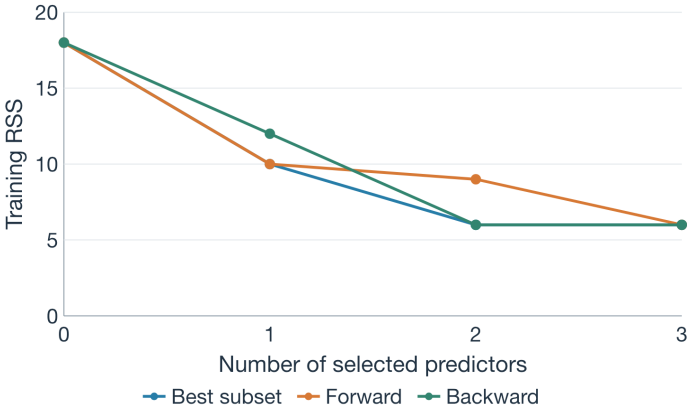
Current model	Possible smaller models	Decision
{1, 2, 3}	{1, 2}: 6 {1, 3}: 9 {2, 3}: 9	Keep {1, 2}
{1, 2}	{1}: 12 {2}: 12	Keep {1}*
{1}	None: 18	Null model

Best singleton: {3}, RSS = 10. Backward singleton: {1}, RSS = 12.

A useful variable may have been removed at an earlier stage.

Entries show training RSS. *Ties keep the smaller predictor index.

Search paths and their consequences



Best subset
Every combination is tried.

Stepwise
Cheaper, restricted paths.

A better training search does not guarantee better prediction.

Cross-validation chooses the size

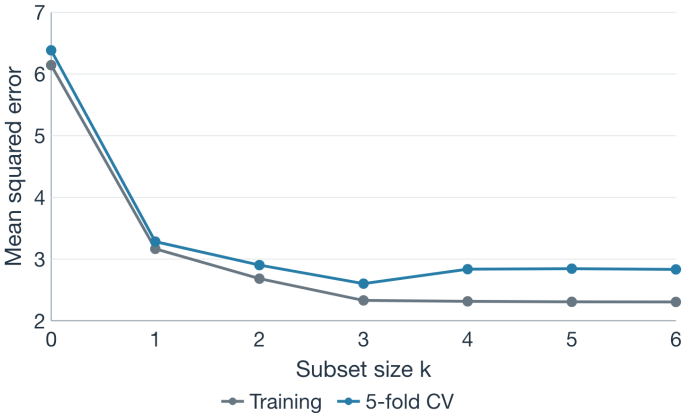
For every candidate size k and each validation fold:

1. Training rows	Run the chosen search method on these rows only; keep the model it gives at size k.
2. Held-out rows	Predict with that fitted subset; record squared errors
3. All folds	Pool the held-out errors; average them to obtain CV MSE

Different folds may select different variables at the same k.

Selecting variables once on all rows leaks validation information into the search.

A larger model can fit better and predict worse



60 observations
6 candidate predictors

CV minimum: k = 3
CV MSE ≈ 2.60

Then search again at k = 3 on all development data. Final subset here: {1, 2, 3}.

Simulated example. The selected CV minimum is not an independent test-error estimate.

Subset selection controls the number of coefficients

$$\min_{\beta_0, \beta} \text{RSS} \quad \text{subject to} \quad \|\beta\|_0 \leq k$$

The “L0 norm” counts nonzero slopes; it is not a true norm.

Once selected, a coefficient can still become very large.
Correlated predictors can make these estimates unstable.

Shrinkage also controls coefficient magnitude, not only inclusion.

This constrained optimization describes best subset selection; stepwise methods approximate it.

Ridge regression

$$\min_{\beta_0, \beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - \beta_0 - x_i^T \beta)^2 + \frac{\lambda}{2} \sum_{j=1}^{p} \beta_j^2 \right\}$$

Fit term	Keep predictions close to the observed responses
Penalty term	Charge more for large slope coefficients
Tuning parameter λ	Controls the trade-off; $\lambda \geq 0$

$\lambda = 0$ gives least squares. The intercept is not penalized. [1-3]

Why predictor scale matters

The same effect: 2 units per meter = 0.02 units per centimeter.

Same predictions, but very different squared-coefficient penalties.

$$x_{ij}^{*} = \frac{x_{ij} - \overline{x}_j}{s_j}, \qquad s_j^2 = \frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2$$

Now each predictor has mean 0 and mean squared value 1.
A coefficient describes a one-standard-deviation change.

Drop constant predictors. During CV, estimate means and scales from the training fold only.

The ridge solution

Center y and standardize X: the intercept is zero in these coordinates.
X is the n × p predictor matrix; y is the n-vector of centered responses.

$$(X^T X + n\lambda I)\widehat{\beta} = X^T y$$

$$\widehat{\beta}^{\text{ridge}} = (X^T X + n\lambda I)^{-1} X^T y$$

I is the p × p identity matrix. For λ > 0, the inverse exists.

For new x, reuse training means/scales. Add back the original training response mean.

Ridge remains well-defined with correlated columns or p > n. Derivation: B1.

A transparent special case: orthogonal predictors

$$X^T X/n = I, \qquad z_j = \frac{x_j^T y}{n} = \widehat{\beta}_j^{\text{OLS}}$$

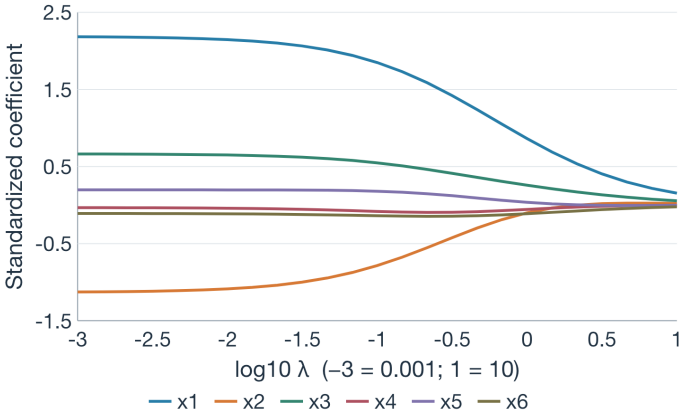
Predictors are uncorrelated in this sample and standardized.
Their coefficient problems separate:

$$\min_{\beta_j} \left\{ \tfrac{1}{2}(\beta_j - z_j)^2 + \tfrac{\lambda}{2}\beta_j^2 \right\}$$

$$(\beta_j - z_j) + \lambda\beta_j = 0 \quad \Longrightarrow \quad \widehat{\beta}_j = \frac{z_j}{1+\lambda}$$

Ridge pulls each OLS coefficient toward zero by the same factor in this special case.

Ridge coefficients along a regularization path



Same 60-observation example as before.

As λ becomes large, all slopes approach zero.

With correlated inputs, a single path need not shrink monotonically.

A regularization path refits the same method over a range of penalty strengths.

Ridge shrinks, but usually does not select

For orthogonal standardized predictors, take these OLS estimates:

Coefficient	1	2	3
OLS	3	1	0.2
Ridge, $\lambda = 1$	1.5	0.5	0.1

Even the weakest estimate remains in the fitted model.

For both shrinkage and exact zeros, change the penalty.

The lasso

$$\min_{\beta_0, \beta} \left\{ \frac{\text{RSS}}{2n} + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

Absolute values replace squared coefficients.
The penalty has a sharp corner at zero.

Some fitted coefficients become exactly zero.
The remaining coefficients are usually shrunk.

The lasso combines coefficient shrinkage and variable selection. [1,2,4]

Why the lasso produces exact zeros

For an orthogonal standardized predictor, solve one coefficient at a time.

$$\min_{\beta} \left\{ \tfrac{1}{2}(\beta - z)^2 + \lambda |\beta| \right\}$$

OLS estimate z	Lasso estimate	Reason
$z > \lambda$	$z - \lambda$	Positive-side derivative is zero
$-\lambda \le z \le \lambda$	0	Left slope ≤ 0 ≤ right slope
$z < -\lambda$	$z + \lambda$	Negative-side derivative is zero

At $\beta = 0$: left slope = $-z - \lambda$; right slope = $-z + \lambda$.
Zero is optimal exactly when $-\lambda \le z \le \lambda$.

Three ways to modify the same OLS estimates

Orthogonal standardized predictors; OLS estimates (3, 1, 0.2).

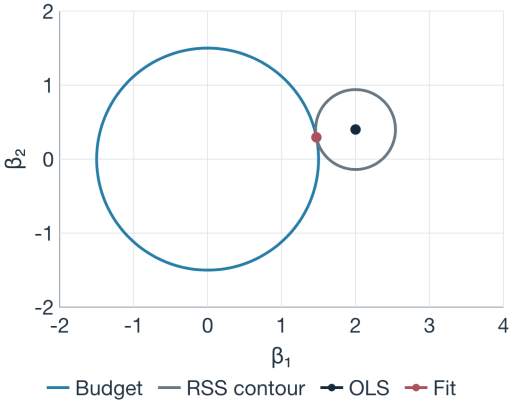
Method	Fitted coefficients	Operation
Best subset, k = 2	(3, 1, 0)	Keep two; refit without shrinkage
Ridge, λ = 1	(1.5, 0.5, 0.1)	Multiply all by 1/2
Lasso, λ = 1	(2, 0, 0)	Soft threshold at 1

Illustrative settings. Equal numeric λ across penalties does not mean equal model complexity.

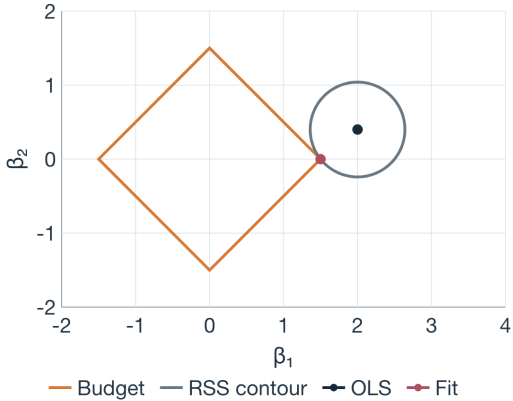
Constraint geometry explains the difference

A suitable coefficient budget gives the same fit as a penalty (see B3).
Expand an RSS contour from OLS; its first contact with the budget gives the fit.

Ridge: $\beta_1^2 + \beta_2^2 \le 1.5^2$



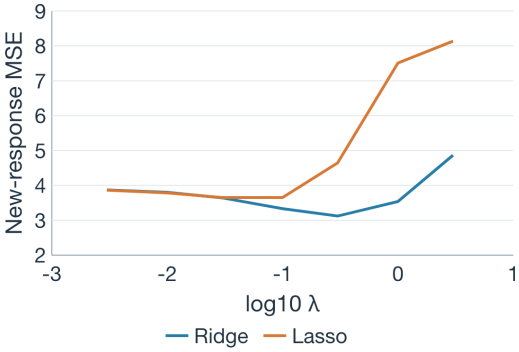
Lasso: $|\beta_1| + |\beta_2| \le 1.5$



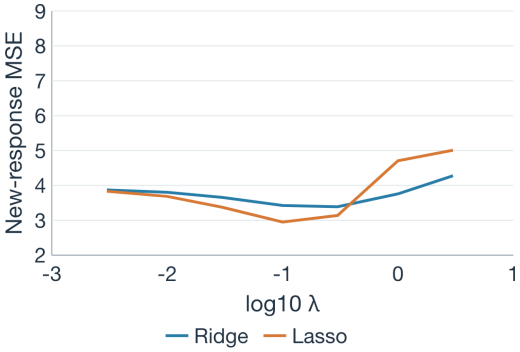
Shown: orthogonal predictors. The lasso fit touches a corner where $\beta_2 = 0$.

Dense and sparse signals can favor different penalties

Dense: 20 nonzero slopes



Sparse: 3 nonzero slopes



100 simulated training sets per scenario; n = 50, p = 20.

Illustrative fixed-λ curves, not CV-tuned performance. Signal strengths also differ; details: B5.

Elastic net combines the two penalties

$$\min_{\beta_0, \beta} \left\{ \frac{\text{RSS}}{2n} + \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right] \right\}$$

Mixing parameter	Penalty
$\alpha = 0$	Ridge
$\alpha = 1$	Lasso
$0 < \alpha < 1$	Lasso sparsity + a stabilizing quadratic term

λ controls overall strength. α controls the mixture. Both can be chosen by CV. [5,6]

Gaussian errors explain the least-squares fit term

$$y_i \mid x_i, \beta_0, \beta \sim N(\beta_0 + x_i^T \beta, \sigma^2)$$

Independent observations; common noise variance σ^2.

$$-\log L(\beta_0, \beta) = \frac{\text{RSS}}{2\sigma^2} + C$$

Maximize likelihood = minimize RSS = ordinary least squares.

A shrinkage penalty needs an additional assumption about coefficients.

A Gaussian prior gives ridge MAP

Prior belief: large standardized slopes are less plausible.
Independently, each slope has a $N(0, \tau^2)$ prior.

$$p(\beta \mid y, X) \propto L(\beta)\, p(\beta)$$

$$-\log p(\beta \mid y, X) = \frac{\text{RSS}}{2\sigma^2} + \frac{\|\beta\|_2^2}{2\tau^2} + C$$

MAP: the coefficient vector with the highest posterior density.

$\lambda = \frac{\sigma^2}{n\tau^2}$ in our ridge objective

Smaller prior variance τ^2 gives stronger shrinkage. Flat prior on the intercept; σ^2 fixed.

A Laplace prior gives lasso MAP

Prior belief: slopes concentrate near zero, with heavier tails than Gaussian.

$$p(\beta_j) = \frac{1}{2b}\exp(-|\beta_j|/b), \qquad b > 0$$

$$-\log p(\beta \mid y, X) = \frac{\text{RSS}}{2\sigma^2} + \frac{\|\beta\|_1}{b} + C$$

$\lambda = \frac{\sigma^2}{nb}$ in our lasso objective

Ridge and lasso are MAP estimates here; ordinary MLE gives OLS.

Independent slope priors; flat intercept prior; σ^2 fixed. More detail: B4.

Choosing among subset selection and shrinkage

Method	What is controlled?	Typical consequence
Best / stepwise subset	Number of predictors	Zeros; other slopes unshrunk
Ridge	Squared slope magnitude	Stable fit; usually dense
Lasso	Absolute slope magnitude	Shrinkage with exact zeros
Elastic net	L1 and L2 together	Sparsity with added stability

Use prediction evidence and the need for a small model together.

Selected variables are not automatically causal or the true support. No method always wins. [7]

A1 · Why RSS falls and search cost rises

Nested feasible sets

$$S \subseteq T \quad \Longrightarrow \quad \min_{\beta_T} \mathrm{RSS}(T) \leq \min_{\beta_S} \mathrm{RSS}(S)$$

Set every added coefficient to zero to reproduce the smaller fit.

Number of fits along a full forward path

$$1 + p + (p-1) + \cdots + 1 = 1 + \frac{p(p+1)}{2}$$

Best subset fits 2^p candidates. These counts include the null model and ignore CV repetition.

A2 · Verifying the three-predictor example

Take four centered orthonormal vectors u, v, w, r in eight dimensions.

$$x_1 = u, \quad x_2 = v, \quad x_3 = (u + v + w)/\sqrt{3}, \quad y = \sqrt{6}\,(u + v + r)$$

The response has squared length 18. For any subset S:

$$\mathrm{RSS}(S) = 18 - c_S^T G_{SS}^{-1} c_S, \quad G = X^T X, \quad c = X^T y$$

$$G_{11} = G_{22} = G_{33} = 1, \quad G_{12} = 0, \quad G_{13} = G_{23} = 1/\sqrt{3}$$

$$c = (\sqrt{6}, \sqrt{6}, 2\sqrt{2})^T \quad \Longrightarrow \quad \mathrm{RSS}(\{3\}) = 18 - (2\sqrt{2})^2 = 10$$

GSS is the submatrix for the included predictors; cS is the corresponding subvector.

A3 · Deriving the Gaussian AIC / BIC form

$$\ell(\beta, \sigma^2) = -\tfrac{n}{2}\log(2\pi\sigma^2) - \frac{\mathrm{RSS}(\beta)}{2\sigma^2}$$

First minimize RSS over coefficients. Then maximize over σ²:

$$\widehat{\sigma}^2 = \mathrm{RSS}/n, \qquad -2\hat{\ell} = n\log(\mathrm{RSS}/n) + C_n$$

There are d = k + 2 parameters: k slopes, one intercept, one variance.

$$\mathrm{AIC} = n\log(\mathrm{RSS}/n) + 2k + C_A$$

$$\mathrm{BIC} = n\log(\mathrm{RSS}/n) + k\log n + C_B$$

A4 · Other size-adjusted fit measures

Mallows' Cp: lower is preferred

$$C_p = \frac{\mathrm{RSS}_k}{\widehat{\sigma}^2} - n + 2(k+1)$$

Use one common noise-variance estimate across the candidate models.
Often estimated from a suitably rich model; reliability matters.

Adjusted R^2: higher is preferred

$$R^2_{\text{adj}} = 1 - \frac{\mathrm{RSS}_k/(n-k-1)}{\mathrm{TSS}/(n-1)}$$

Residual degrees of freedom n − k − 1 account for k slopes and the intercept.

B1 · Ridge: gradient and uniqueness

$$J(\beta) = \frac{1}{2n}(y - X\beta)^T(y - X\beta) + \frac{\lambda}{2}\beta^T\beta$$

$$\nabla J = -\frac{X^T(y - X\beta)}{n} + \lambda\beta = 0$$

$$(X^TX + n\lambda I)\widehat{\beta} = X^Ty$$

For any nonzero vector a and λ > 0:

$$a^T(X^TX + n\lambda I)a = \|Xa\|_2^2 + n\lambda\|a\|_2^2 > 0$$

$$\beta_j^{\text{raw}} = \widehat{\beta}_j/s_j, \qquad \beta_0^{\text{raw}} = \overline{y} - \sum_j \overline{x}_j\beta_j^{\text{raw}}$$

B2 · Soft thresholding, including the zero case

$$X^T X/n = I \quad \Longrightarrow \quad \frac{\text{RSS}}{2n} = C + \frac{1}{2}\sum_j (\beta_j - z_j)^2$$

At β = 0, the supporting-line slopes (subgradients) of |β| fill [−1, 1].

$$0 \in -z + \lambda[-1, 1] \quad \Longleftrightarrow \quad |z| \le \lambda$$

Away from zero: β − z + λ sign(β) = 0.

$$\widehat{\beta} = \text{sign}(z)(|z| - \lambda)_+, \qquad (a)_+ = \max(a, 0)$$

With correlated predictors, the joint lasso solution is not simply thresholded OLS.

$$\min_{\beta} \frac{\mathrm{RSS}}{2n} \quad \text{subject to} \quad P(\beta) \le t$$

$$\mathcal{L}(\beta, \lambda) = \frac{\mathrm{RSS}}{2n} + \lambda(P(\beta) - t), \quad \lambda \ge 0$$

For fixed λ, the term −λt is constant in β.
Appropriate constraints and penalties share solutions for ridge / lasso.

Under Gaussian noise, a fixed coefficient constraint gives a constrained MLE. A prior instead gives the MAP interpretation.

λ values depend on factors such as 1/n and 1/2. Never compare λ across different conventions.

B4 · Matching prior scales to our penalties

Multiply the negative log-posterior by σ^2/n; the minimizer is unchanged.

Independent slope prior	Penalty after rescaling	Parameter match
Gaussian $N(0, \tau^2)$	$\frac{\sigma^2}{2n\tau^2}\|\beta\|_2^2$	$\lambda = \sigma^2 / (n\tau^2)$
Laplace scale b	$\frac{\sigma^2}{nb}\|\beta\|_1$	$\lambda = \sigma^2 / (nb)$

MAP is a posterior mode, not generally a posterior mean.
A continuous Laplace prior has no point mass at zero;
its sharp peak can still make the posterior mode exactly zero.

B5 · How the numerical illustrations were generated

Slides 19 and 25: one simulated data set

n = 60; p = 6; β = (2, −1.5, 1, 0, 0, 0); noise SD = 1.5.
Mean-zero Gaussian predictors: Cov(xj, xk) = 0.4^|j−k|. Seed 4432.
Five-fold subset CV: variable selection repeated within each training fold.

Slide 31: 100 independent training sets per scenario

n = 50; p = 20; mean-zero Gaussian predictors.
Cov(xj, xk) = 0.3^|j−k|; noise SD = 1.5.
Dense: all slopes 0.4. Sparse: (1.4, −1.2, 0.8, 0, ..., 0).
Average exact population MSE over 100 fits at each fixed λ.

$$\text{Risk} = \sigma^2 + (\widehat{\beta} - \beta)^T \Sigma (\widehat{\beta} - \beta) + \widehat{\beta}_0^2$$

Original-scale coefficients; zero true intercept. These scenarios also differ in signal strength.

References · Concepts and derivations

[1] James, Witten, Hastie & Tibshirani (2021)
An Introduction to Statistical Learning, 2nd ed. Chapter 6.

[2] Hastie, Tibshirani & Friedman (2009)
The Elements of Statistical Learning, 2nd ed. Sections 3.3-3.4, 7.

[3] Hoerl & Kennard (1970)
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[4] Tibshirani (1996)
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[5] Zou & Hastie (2005)

Regularization and Variable Selection via the Elastic Net.

[6] Friedman, Hastie & Tibshirani (2010)

Regularization Paths for Generalized Linear Models
via Coordinate Descent.

[7] Hastie, Tibshirani & Tibshirani (2020)

Best Subset, Forward Stepwise or Lasso?
Analysis and Recommendations Based on Extensive Comparisons.